

Topic: Hotel Room Booking & Management Database

Description

The Hotel Room Booking and Management Database is a centralized relational database system designed to streamline day-to-day operations for a modern hospitality and lodging business. Managing a hotel efficiently requires tracking multiple dynamic variables, including real-time room availability, guest verifications, check-in timelines, and transaction records. This project implements a structured database architecture that ensures data consistency, eliminates redundancy, and optimizes operational efficiency across the property.

At its core, the schema organizes data into three critical business segments: inventory management (Rooms), guest records (Guests), and transactional booking operations (Reservations). It models real-world logical constraints, such as ensuring a room cannot be double-booked for overlapping dates, maintaining strict unique identity verification for guest compliance, and tracking pricing variations.

By progressing through the full database lifecycle, this project demonstrates end-to-end relational database management capabilities. It moves from initial requirement analysis and entity-relationship mapping to schema implementation and structural optimization. Advanced database features can be used to automate recurring tasks like reservation cost computations based on stay duration, while triggers can dynamically flag room availability statuses to preserve data integrity. Ultimately, the system serves as a scalable foundation for business analytics, tracking room occupancy rates and revenue metrics to maximize hospitality profitability.

Entities Required

Entity 1: Rooms

Entity 2: Guests

Entity 3: Reservations

Data Structure Tables

1. Rooms Entity

Table	Attributes
Rooms	room_id (Unique Identifier / Primary Key) • room_number (Must be unique, e.g., '101', '202') • room_type (e.g., 'Standard', 'Deluxe', 'Suite') • price_per_night (The base rental price per day) • status (Available, Booked, Being Cleaned)

2. Guests Entity

Table	Attributes
Guests	guest_id (Unique Identifier / Primary Key) • name (Full name of the visitor) • email (Must be unique for communication) • phone (Contact number) • government_id_number (Crucial for legal hotel check-in compliance)

3. Reservations Entity

Table	Attributes

Reservations	reservation_id (Unique Identifier / Primary Key) • guest_id (Links back to the Guests table) • room_id (Links back to the Rooms table) • check_in_date (Start date of stay) • check_out_date (End date of stay) • total_amount (Total cost computed for the stay duration)
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